

A Quasi-Static Delamination Model with Rate-Dependent Interface Damage Exposed to Cyclic Loading

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Abstract. A model for numerical analysis of interface damage which leads to interface crack initiation and propagation in multi-domain structures under cyclic loading is considered. Modelling of damage takes into account various relations between interface stresses and displacement gaps providing the response of a cohesive zone model, additionally equipped by a kind of viscosity associated to the evolution of the interface damage. Together with repeating loading-unloading conditions, it makes this damage process to have a fatigue-like character, where the crack appears for smaller magnitude of the cyclic load than for pure uploading.

Introduction

Models which are able to solve various civil engineering problems with rupturing interfaces are highly demanded. In this paper, we intend to model damage evolution under loading with a constant amplitude which may cause total damage for the amplitudes substantially smaller than the load in pure uploading. The simulation of fatigue processes with cohesive zone models (CZM) was introduced by their cyclic variant in [1, 2, 3].

The present approach is physically based on energy evolution, which includes stored elastic energy, energy of the external load and also dissipation due to damage. The dissipation potential has a rate dependent form with included viscosity associated to damage, see also [4, 5]. The CZMs in a context of the energy evolution were implemented in [6]. Here, a viscous term was added to the energy dissipation term in order to provide a model in which the unloading and reloading paths are uncoincident. Other requirements were to decrease transferable force with increasing damage and to maintain a lower bound of stress for damage evolution.

The solution evolution is approximated by a semi-implicit time stepping algorithm (a staggered scheme) providing a variational structure to the solved problem by two recursively solved minimisations with respect to separated deformation and damage variables, described in the previous author's works [6, 7, 8, 9]. This also comprehended the Symmetric Galerkin BEM (SGBEM) [10, 11] to provide the displacement and stress solutions for the solids and (sequential-)quadratic programming algorithms to calculate the solutions for energy minimisations in a variationally based approach [12].

The presented numerical results demonstrate the properties of the proposed model in a very simplified configuration and reveal a characteristic response of the model to cyclic loading.

Description of the Model

Let us consider the energies which are taken into account in the model for two domains Ω^A and Ω^B with boundaries Γ^A and Γ^B , respectively. The stored energy functional is defined as

$$E(t; u, \zeta) = \sum_{\eta=A,B} \int_{\Gamma_\eta} \frac{1}{2} u^\eta \cdot p^\eta \, d\Gamma + \int_{\Gamma_c} \frac{1}{2} \left[k_n \Phi(\zeta) \llbracket u \rrbracket_n^2 + k_t \Phi(\zeta) \llbracket u \rrbracket_t^2 + k_g \left(\llbracket u \rrbracket_n^- \right)^2 \right] \, d\Gamma \quad (1)$$

for the admissible (time t dependent) displacement $u^\eta = g^\eta(t)$ on a part of boundary Γ_η with $\eta = A$, or B , and where p^η denotes pertinent surface tractions. The first integral, representing the elastic strain

energy in the subdomains Ω^η , is expressed in its boundary form. The last integral is the interface term which defines dependence of the interface stiffnesses k_n and k_t , on the actual state of damage by the function $\Phi(\zeta)$, so that $\zeta = 1$ pertains to the initial undamaged state and $\zeta = 0$ means total damage (a crack). The terms $[[\cdot]]_n$ and $[[\cdot]]_t$, respectively, are the normal and tangential relative displacements between opposite interface points, i.e. $[[u]] = u^A - u^B$ (superscripts A and B refer to the bodies adjacent to the interface from both its sides). The last term with $[[\cdot]]_n^-$, denoting the negative part of the relative normal displacement, reflects normal-compliance contact condition being understood as a reasonable approximation of Signorini contact conditions.

The potential energy of external forces (acting only along a part of the boundary denoted Γ_N^η) is given by the relation

$$W(t, u) = - \sum_{\eta=A, B} \int_{\Gamma_N^\eta} f_N^\eta(t) \cdot u^\eta d\Gamma, \quad (2)$$

where f_N^η are the prescribed tractions on the part of boundary Γ_N^η .

The dissipated energy is introduced by the (pseudo)potential which reflects the rate-dependence and unidirectionality of the debonding process

$$R(\dot{\zeta}) = \int_{\Gamma_c} -\alpha_1 \dot{\zeta} + \frac{1}{2} \alpha_2 \dot{\zeta}^2 d\Gamma, \quad \text{for } \dot{\zeta} \leq 0, \quad (3)$$

where α_1 is related to the instant of damage initiation and α_2 is visco-damage parameter.

The relations which govern the evolution of damage can be written in form of nonlinear variational inclusions with initial conditions

$$\begin{aligned} \partial_u E(t; u, \zeta) + \partial_u W(t; u) \ni 0, \quad u|_{t=0} = u_0, \\ \partial_{\dot{\zeta}} R(\dot{\zeta}) + \partial_{\zeta} E(t; u, \zeta) \ni 0, \quad \zeta|_{t=0} = \zeta_0, \end{aligned} \quad (4)$$

where ∂ denotes (partial) subdifferential of a convex non-smooth function which can be replaced by Gateaux differential if the functional is sufficiently smooth, as e.g. F . The initial condition for damage is usually $\zeta_0 = 1$, pertaining to the undamaged state. The energy release rate is obtained from the second inclusion of Eq. 4 as it expresses a change of energy E due to a change of damage ζ .

A Simplified Model

The model provides some hysteresis during one cycle of the load and leads to a crack in an element which could bear larger load if not applied in cycles. This is documented first in the following very simplified situation.

The interface in the model can be seen as a series of interface springs which dissipates energy in the process of damage. In what follows, we intend to consider one such a ‘spring’ near the crack tip, cf. [6, 7]. A simplified model for the ‘spring’ under consideration includes energy from Eq. 1 and dissipation from Eq.3 in the form

$$E = \frac{1}{2} k \zeta u^2, \quad R = -\alpha_1 \dot{\zeta} + \frac{1}{2} \alpha_2 \dot{\zeta}^2 \quad (5)$$

where we consider the simplest choice of the function Φ : $\Phi(\zeta) = \zeta$. We also suppose that the spring is loaded by a cyclic displacement load

$$u(t) = v |t - 2nt_m|, \quad \text{for } t \in [(2n-1)t_m; (2n+1)t_m], \quad n = 0, 1, 2, \dots, t > 0, \quad (6)$$

so that $u_m = vt_m$ is the amplitude. The system Eq. 4 then contains only the second equation which reduces to the complementarity conditions for $\zeta \in [0; 1]$

$$\alpha_1 - \frac{1}{2} k u^2(t) - \alpha_2 \dot{\zeta} \geq 0, \quad \dot{\zeta} \leq 0, \quad \left(\alpha_1 - \frac{1}{2} k u^2(t) - \alpha_2 \dot{\zeta} \right) \dot{\zeta} = 0. \quad (7)$$

This means that if we choose t_m as $t_m \leq t_0$ with $t_0 = \frac{1}{v} \sqrt{\frac{2g}{k}}$, the damage is not initiated because $\alpha_1 \geq \frac{1}{2}ku^2(t)$ for any t .

Similarly, for $t_m > t_0$ and $t \in [2nt_m - t_0; 2nt_m + t_0]$, damage does not evolve, i.e. $\dot{\zeta} = 0$, so that if we denote $\zeta(2nt_m) = \zeta_n$ (on total unloading), for all these t we have $\zeta(t) = \zeta_n$. Assuming $t \in [2(n-1)t_m + t_0; 2nt_m - t_0]$, the damage satisfies the differential equation

$$\dot{\zeta} = \frac{1}{\alpha_2} \left(\alpha_1 - \frac{1}{2}ku^2(t) \right), \quad \zeta(2(n-1)t_m + t_0) = \zeta_{n-1} \quad (8)$$

which provides also $\zeta_n = \zeta(2nt_m - t_0)$. The solution obtained by mere integration is

$$\zeta(t) = \begin{cases} \zeta_{n-1} + \frac{\alpha_1}{\alpha_2} (t - 2(n-1)t_m - t_0) - \frac{kv^2}{6\alpha_2} ((t - 2(n-1)t_m)^3 - t_0^3), \\ \text{for } t \in [2(n-1)t_m + t_0; (2n-1)t_m], \\ \zeta((2n-1)t_m) + \frac{\alpha_1}{\alpha_2} (t - (2n-1)t_m) + \frac{kv^2}{6\alpha_2} (((2n-1)t_m - t)^3 - t_m^3), \\ \text{for } t \in [(2n-1)t_m; 2nt_m - t_0], \end{cases} \quad (9)$$

rendering a recursive relation

$$\zeta_n = \zeta_{n-1} - \frac{(2\alpha_1)^{\frac{3}{2}}}{3\alpha_2 vk^{\frac{1}{2}}} \left(\left(\frac{t_m}{t_0} \right)^3 - 3\frac{t_m}{t_0} + 2 \right), \quad \zeta_0 = 1. \quad (10)$$

The constant difference $\Delta\zeta = \zeta_n - \zeta_{n-1}$ is a change of ζ during one cycle of the load. Thus, we can calculate the number N of cycles to total damage as $N = -\frac{1}{\Delta\zeta}$. Knowing that, it is possible to plot the Wöhler diagram pertinent to present model as in Fig.1. For this particular plotting we chose:

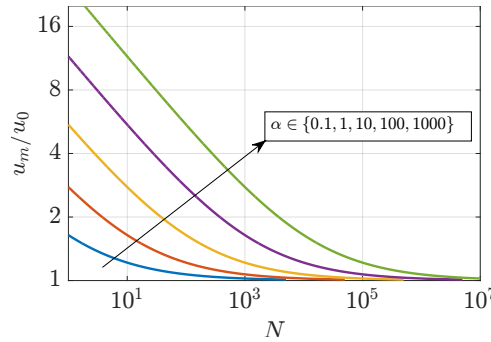


Fig. 1: The Wöhler diagram for the simplified one-spring model, the influence of the parameter α_2 .

$\alpha_1 = 1\text{Jm}^{-2}$, $k = 2\text{MPam}^{-1}$, $v = 10\text{mms}^{-1}$ and various α_2 [Jsm^{-2}] to demonstrate the influence of the viscous term which causes the particular damage evolution with a fatigue character. We can also observe an endurance limit determined by the value u_0 . Here, u_0 is the maximal displacement which does not cause any damage, i.e. $u_0 = vt_0$, or p_0 is pertinent traction applied to the spring, i.e. $p_0 = ku_0$.

Additionally, the traction $p : p(t) = k\zeta(t)u(t)$ with substituted Eqs. 6 and 9 applied to the spring provides a stress-displacement diagram for the spring under the cyclic load, see Fig. 2. In the graphs, various amplitudes of the applied displacement load u_m are used. The number of the cycles needed to totally damage the spring can also be checked by the corresponding curve in Fig. 1. The graph of damage evolution also shows uniform damage decrease through all the life of the spring. This is however only a feature of this particular simplified model whose endurance zone is influenced by the choice of $\Phi(\zeta)$.

Similar graphs could be obtained if the spring is loaded by a cyclic force. Then, energy of external forces W from Eq. 2 should be added to the functions E and R in Eq. 5 and $u(t)$ in Eq. 6 is replaced by time dependent applied stress, namely

$$W = -pu, \quad \text{with } p(t) = v_p |t - 2nt_m| \quad (11)$$

for t from the same range as introduced in Eq. 6, correspondingly $p_m = v_p t_m$ is the amplitude.

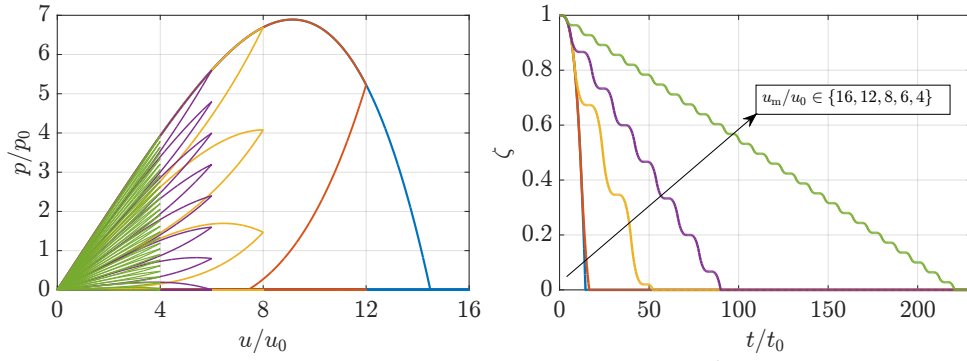


Fig. 2: The working diagram and evolution of ζ for $\alpha_2 = 100 \text{Jsm}^{-2}$, influence of the amplitude u_m .

The obtained solution for the same values of the parameters and for $v_p = kv$ is shown in Fig. 3. Here, the endurance limit is p_0 . The change of ζ during one cycle of the load $\Delta\zeta$ is not constant now.

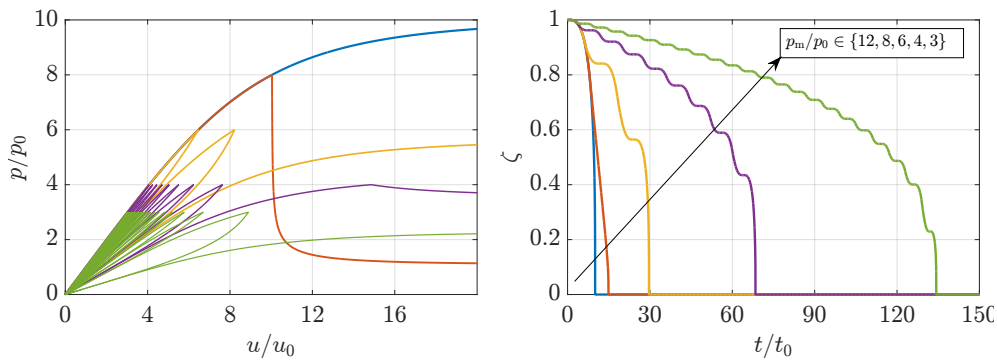


Fig. 3: The working diagram and evolution of ζ for $\alpha_2 = 100 \text{Jsm}^{-2}$, influence of the amplitude p_m .

It increases in time making possible abrupt damage changes and expectedly fast crack propagation in the interface model of Eqs. 1 to 4.

A Beam under a Cyclic Load

We use the model to find solutions for various amplitudes of the same load applied to the beam shown in Fig.4, see also [6, 8]. The numerical procedures proposed for the solution of the problem Eq. 4 con-

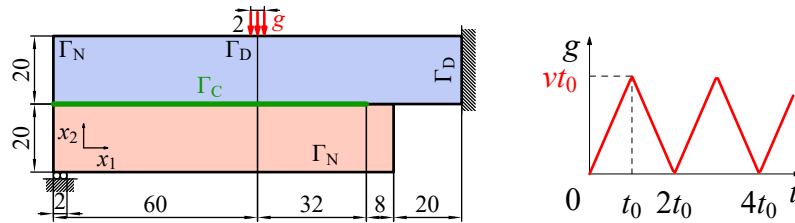


Fig. 4: The beam with its supports and time dependent displacement load.

sider the time discretization and the spatial discretization separately. The time discretization scheme is defined by a semi-implicit algorithm with a fixed time step size τ and introducing the time instants t^n . In order to obtain such an algorithm from Eq. 4, the damage rate is approximated by the finite difference $\dot{\zeta} \approx (\zeta^n - \zeta^{n-1}) / \tau$, where ζ^n denotes the solution at the instant t^n . For detail, see e.g. [6]. The role of the SGBEM in the present computational procedure is to provide a complete boundary-value solution from the given boundary data for each domain in order to calculate the elastic strain energy in these domains. Thus, the SGBEM code calculates also unknown tractions along the interface, assuming the displacements gaps to be known from the used minimization procedure, in the same way as proposed and tested in [7, 10]. The model formulation is implemented in a MATLAB computer code.

Table 1: Number of time steps Δn to reach crack length of $a = 14\text{mm}$, $\tau = 5\text{ms}$.

g_m [mm]	0.025	0.05	0.1	0.2
$n_{a=0\text{mm}}$	1494	352	95	33
$n_{a=14\text{mm}}$	4425	972	230	49
Δn	2931	620	135	16

The material properties of both layers in our example are: $E = 70\text{GPa}$, $\nu = 0.35$. The beam is loaded by a time dependent displacement load with the amplitude $g_m = vt_0$. It is also shown in Fig. 4, where $v = 1\text{mm s}^{-1}$ and for t_0 are used various values. The characteristics of the interface needed for functionals in Eqs. 1 and 3 are $\alpha_1 = 1.01\text{Jm}^{-2}$, $\alpha_2 = 100\text{Jsm}^{-2}$, $k_n = 2k_t = 2\text{TPam}^{-1}$ and $\Phi(\zeta) = 100\zeta/(101 - \zeta)$, cf. [8].

The discretization is uniform such that the BE mesh size (element length) is $h = 2\text{mm}$ and the pertinent time step is $\tau = 5\text{ms}$. We had solved the problem with this mesh until we reached a crack of the length $a = 14\text{mm}$. Table 1 summarises number of time steps $n_{a=0\text{mm}}$ needed to initiate a crack (in the sense that ζ at the right end of the interface falls below 0.1) and number of time steps $n_{a=14\text{mm}}$ for which the condition is satisfied at the point 14mm far from the end point. With given τ , the number of cycles ΔN which leads to supposed crack length is $\Delta N = \Delta n \cdot \tau / (2t_0)$. By linearised regression, we found that $\Delta a / \Delta N = 7876.2 \cdot g_m^{3.4751}$ ($R^2 = 0.9964$), which is a good approximation of the Paris power law of fatigue crack growth, [13].

The relations that can be check at the interface where a crack propagates include traction-separation relation and evolution of damage. Both of them are shown in Fig. 5 for two distinct interface points which are, respectively: a point close to the end point of the interface where the damage (and subsequently crack) is initiated and the point at the crack tip if $a = 14\text{mm}$ as above. The graphs are similar to

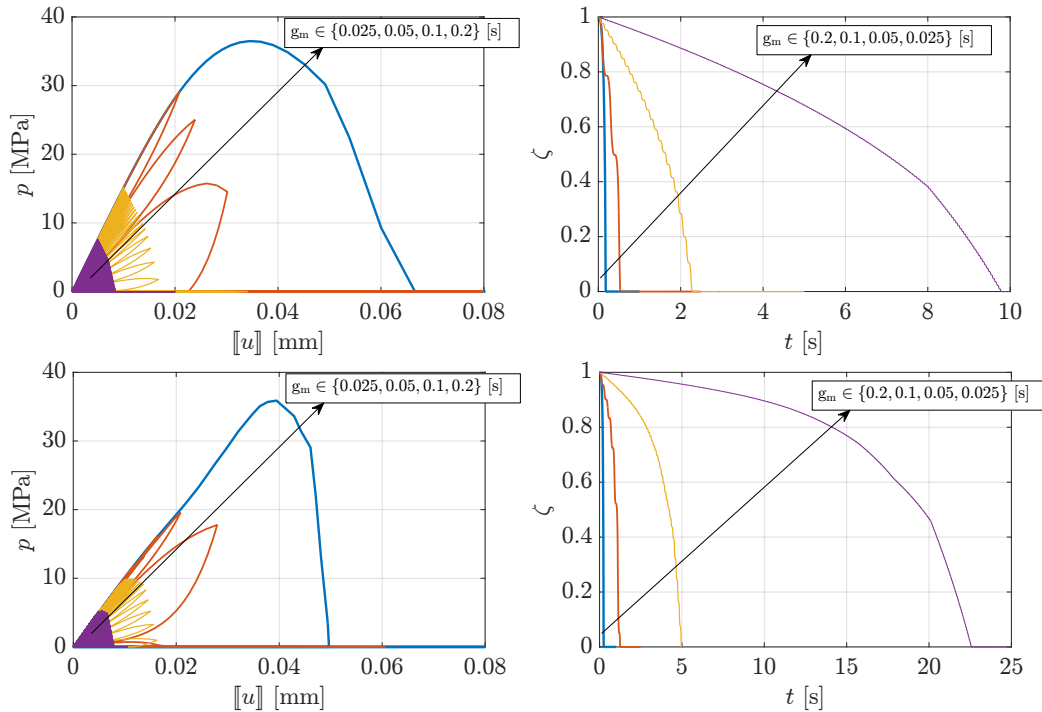


Fig. 5: The stress-displacement-gap relation and evolution of ζ for a near-to-the-end point at the interface $x_1 = 90\text{mm}$ (top), and for the point at referenced crack-length $a = 14\text{mm}$, i.e. $x_1 = 78\text{mm}$ (bottom), influence of the amplitude g_m .

those shown in Figs. 2 and 3 for the very simple configuration. Though here, influenced by the neighbourhood of a particular point, the decrease of the stress appear to be faster as the crack tip approaches that point.

Conclusion

A model for solving interface damage previously presented in [6] was enhanced here to include some hysteresis in damage to observe its behaviour under cyclic loading. The presented results confirm that the model is capable to evolve the damage in the sense of fatigue life. Of course, there are many parameters in the model that should be appropriately set to follow satisfactorily some experimental results. Nevertheless, it might be expected that the proposed approach turns to be useful in practical engineering calculations.

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